Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years developing scalable, testable applications using **Python**, **Java**, React, TypeScript, and SQL. Experienced with Claude, Codex, and Cursor for AI-assisted implementation, testing, debugging, documentation, and code review.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant **Python** platform with Azure SQL and Blob Storage for 500+ users, optimizing relational models, indexed queries, and access paths to reduce application latency from 1.2 seconds to 350 milliseconds.
- Maintained cloud application workflows processing 5,000+ monthly jobs, resolving defects through validation, structured logging, monitoring, recovery procedures, regression testing, and CI/CD-backed releases that sustained greater than 99% production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, using AI-assisted coding tools to implement, review, debug, test, and document maintainable production workflows efficiently.
- Developed a multithreaded processing pipeline for 1,000+ files with validation, fault isolation, progress monitoring, and repeatable recovery, reducing turnaround 60% while preserving stable application behavior under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **AI pipeline** processing 100,000+ records, improving output quality 18% through modular implementation, reproducible experiments, automated validation, performance tracing, and structured debugging across evolving software configurations.
- Implemented spatial-temporal modeling that combined HD-map geometry with vehicle trajectories through multi-head attention, improving closed-loop stability 15% while supporting systematic debugging and comparative analysis across complex scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise integration software with automated builds and VM-based validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Developed object-oriented microservices and AWS data pipelines using Glue, S3, SQL, caching, and pagination, improving response times 30%, processing efficiency 68%, and maintainability across integrated platforms with reliable operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, TypeScript, and Node.js features across eight microservices, integrating WebSockets and REST APIs to synchronize more than two million records across responsive customer and operational workflows.
- Automated testing and cloud delivery with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure monitoring, cutting deployment time 50% while improving release validation and production observability across environments.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS
Applications & APIs: React.js, Next.js, Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, microservices, responsive web applications
Databases: SQL, PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, relational modeling, indexing, query optimization, schema design, data validation
AI Development Tools: Claude, Codex, Cursor, AI-assisted implementation, code review, test generation, debugging, refactoring, documentation, prompt engineering
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux, logging, monitoring
Engineering Practices: Agile, SDLC, OOP, scalable design, unit testing, integration testing, regression testing, bug resolution, code reviews, technical documentation

Projects

Big-Brain | Live | *Python, LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an AI document platform processing 50,000+ records with semantic search, summarization, secure access, and incremental indexing, reducing retrieval latency 40% through optimized embedding and similarity-search pipelines.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

PiXMod | Live | *Next.js, PyTorch, Transformers, MongoDB*
- Built a cloud AI application using Next.js, Transformers, and MongoDB for segmentation, retrieval, and generative editing, processing 1,000+ assets through responsive workflows, scalable metadata storage, and deployment.